Symmetry

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Symmetry Practice: How many lines of symmetry does a regular hexagon have, and what is its order of rotational symmetry?

Answer: A regular hexagon has 6 sides, so it has 6 lines of symmetry and rotational symmetry of order 6.

Symmetry Basics (15 questions)

1. How many lines of symmetry does a regular pentagon have?
2. How many lines of symmetry does an isosceles triangle have (with exactly two equal sides)?
3. How many lines of symmetry does a non-square rhombus have?
4. How many lines of symmetry does a regular octagon have? A) 4 B) 6 C) 7 D) 8 E) 9
5. How many lines of symmetry does the capital letter W have?
6. Which of these capital letters has 2 lines of symmetry? A) F B) X C) N D) S E) L
7. How many lines of symmetry does the capital letter Z have?
8. How many lines of symmetry does a kite have?
9. How many lines of symmetry does a scalene triangle (a triangle with three different side lengths) have? A) 0 B) 1 C) 2 D) 3 E) Infinitely many
10. Select ALL that apply: which of these shapes have exactly 0 lines of symmetry? A) Parallelogram (not a rectangle or rhombus) B) Scalene triangle C) Square D) Letter N E) Letter T
11. Reflect the point (6, 1) in the x-axis. Give the coordinates of the image.
12. Reflect the point (−3, 2) in the y-axis. Give the coordinates of the image.
13. A shape has 1 vertical line of symmetry. If the right-hand half has an area of 22 cm², what is the area of the whole shape?
14. Reflect the point (5, 5) in the x-axis. What are the coordinates of the image? A) (5, −5) B) (−5, 5) C) (−5, −5) D) (5, 5) E) (0, 0)
15. A regular hexagon has how many lines of symmetry?

Applied Symmetry (15 questions)

1. What is the order of rotational symmetry of a regular octagon?
2. What is the order of rotational symmetry of a rhombus that is not a square?
3. What is the order of rotational symmetry of a regular pentagon? A) 1 B) 3 C) 4 D) 5 E) 10
4. What is the order of rotational symmetry of an isosceles trapezium (not a parallelogram)?
5. A regular polygon has 7 lines of symmetry. What is its order of rotational symmetry, and how many sides does it have?
6. Which of these shapes has rotational symmetry of order 2 but NO lines of symmetry? A) Rectangle B) Letter Z C) Square D) Isosceles triangle E) Regular hexagon

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7. Reflect the point (4, −3) in the line y = x. Give the coordinates of the image.

8. Reflect the point (1, 6) in the horizontal line y = 2. Give the coordinates of the image.

9. Select ALL that apply: which of these capital letters have rotational symmetry of order 2? A) O B) Z C) T D) N E) H

10. A shape drawn on a grid has a vertex at (2, 3). It is reflected in the vertical mirror line x = 8 to complete a symmetric figure. Find the coordinates of the image.

11. A rhombus (not a square) has how many lines of symmetry in total? A) 0 B) 1 C) 2 D) 3 E) 4

12. A regular shape has 11 lines of symmetry. How many sides does it have, and what is its order of rotational symmetry?

13. The point (−4, 5) is reflected in the vertical line x = −1. Find the coordinates of the image.

14. Name a shape that has both line symmetry and rotational symmetry of order greater than 1, from: scalene triangle, parallelogram, square, kite.

15. A shape has both line symmetry and rotational symmetry of order greater than 1. Which of these is NOT an example? A) Regular pentagon B) Rectangle C) Rhombus D) Letter N E) Square

Standard / Advanced (15 questions)

1. A regular polygon has rotational symmetry of order 9. How many lines of symmetry does it have?

2. A shape has 3 more lines of symmetry than a square. How many lines of symmetry does it have?

3. The point (2, 9) is reflected in the x-axis, and then the image is reflected again in the x-axis. What are the final coordinates?

4. A regular polygon's order of rotational symmetry is half the number of lines of symmetry of a regular octagon. How many sides does the polygon have?

5. A letter has rotational symmetry of order 2 and no lines of symmetry. Name one capital letter that fits, choosing from: H, N, O, T.

6. A regular polygon has the same number of lines of symmetry as a regular hexagon, but its order of rotational symmetry is double that of a square. Is it possible for a genuine regular polygon to have these two properties at once? Explain, and give a reason.

7. A point P = (3, 7) is reflected in the line x = 5, and the image is then reflected in the line x = 5 again. What are the final coordinates? A) (3, 7) B) (7, 3) C) (−3, 7) D) (3, −7) E) (13, 7)

8. A shape is made from a square (4 lines of symmetry, rotational order 4) joined edge-to-edge with an identical square to form a 2:1 rectangle. How many lines of symmetry does the resulting rectangle have, and what is its order of rotational symmetry?

9. A regular polygon has 2 more lines of symmetry than a regular pentagon. What is its order of rotational symmetry? A) 5 B) 6 C) 7 D) 8 E) 9

10. Point A = (2, 2) is reflected in the line y = x to give point B. Point B is then reflected in the x-axis to give point C. Finally, point C is reflected in the y-axis to give point D. Find the coordinates of D.

11. A regular polygon has rotational symmetry of order n. A second regular polygon has twice as many sides, so its order of rotational symmetry is 2n. If the first polygon is a square, how many lines of symmetry does the second polygon have?

12. Select ALL that apply: a shape is reflected in the line y = x. Which of these statements about the reflection are true? A) The point (0, 0) maps to itself. B) A point on the line y = x always maps to itself. C) The x- and y-coordinates of every point are swapped. D) The point (3, −2) maps to (2, −3). E) The reflection always changes the sign of both coordinates.

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13. A square has side length 8 cm. It is cut along both of its diagonals through the centre into 4 triangles. What type of triangle is each of the 4 pieces, and what is the area of each piece?

14. A regular polygon has rotational symmetry of order n, where n is also its number of lines of symmetry. If $n^2 = 144$, how many sides does the polygon have? A) 10 B) 11 C) 12 D) 13 E) 14

15. A shape has point symmetry (rotational symmetry of order 2, but no lines of symmetry). It is first reflected in the x-axis, and the image is then rotated 180° about the origin. Explain why this combined transformation is the same as a single reflection in the y-axis, and state whether the final image is in the same position as the original shape.

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